

Install Multiple OS in Raspberry Pi for Enhanced Performance



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Raspberry Pi is powerful enough for a lot of applications but we can further enhance its capabilities by simple tweaks. I learnt these tweaks when I wanted to run XBMC Media Center on my Raspberry Pi but also have Raspbian Wheezy for various other things. Off course, I could take the backup using any of the backup tools like win32DiskImager, but I wanted to avoid the hassle of every time writing the image on the SD card or keeping separate SD cards for each OS.

When I successfully installed both the operating systems, I wished I could have a little more power for the XBMC Media Center. I found various ways to overclock Raspberry Pi. Let us see the one that worked for me.

Overclocking Raspberry Pi

Raspberry Pi Foundation has officially allowed overclocking and made it easy for everyone to squeeze more power out of the same Raspberry Pi. There are a few different combinations of frequencies that you can use to boost the CPU and GPU in your Raspberry Pi.

To overclock Raspberry Pi, first of all, check the frequency at which the device is running by writing the command mentioned below in the LxTerminal. You can also do it by running the Raspberry Pi in headless mode (refer article 'Remote Access to Raspberry Pi' published in June issue).

```
$ cat /proc/cpuinfo
```

The other way to see the frequency

```
pi@raspberrypi ~$ cat /proc/cpuinfo
processor       : 0
BogoMIPS      : 38.40
cache_size     : 32K
CPU implementer : 0x41
CPU architecture: 7
CPU part       : 0x0000
CPU revision   : 0

Hardware      : BCM2708
Revision      : 0004
Serial        : 0000000000000000
pi@raspberrypi ~$
```

Fig. 1: CPU Information; the frequency is indicated against 'BogoMIPS'

```
pi@raspberrypi ~$ cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq
700000
pi@raspberrypi ~$
```

Fig. 2: Frequency of Raspberry Pi

```
pi@raspberrypi /boot$ nano config.txt
# Uncomment to increase signal to HDMI, if you have interference, blanking, or
# no display
hdmi_hblank_ignore=1

# Uncomment for composite PAL
#force_pal=1

# Uncomment to overclock the arm. 700 MHz is the default.
arm_freq=800

# For more options see http://elinux.org/RPi_config.txt

pi@raspberrypi /boot$
```

Fig. 3: Editing 'config.txt' file

```
pi@raspberrypi ~$ vopmode get_config arm_freq
arm_freq=800
pi@raspberrypi ~$
```

Fig. 4: Verification of changed frequency

```
pi@raspberrypi ~$ vopmode get_config arm_freq
arm_freq=1000
pi@raspberrypi ~$
```

Fig. 5: Raspberry Pi running at 1 GHz

of Raspberry Pi is as shown below (refer Fig. 2):

```
$ cat /sys/devices/
```

```
system/cpu/cpu0/
cpufreq/scaling_
cur_freq
```

Now that we know the frequency on which Raspberry Pi is running, we can try to overclock it and verify the same. There are a couple of different ways to overclock Raspberry Pi, but the most convenient is to manually edit the configuration file found at '/boot/config.txt'. The best part of this method is that it not only gives you fine control on what parts of the board to overclock, but it also allows you to change other aspects of the process, such as voltage, temperature, thresholds, and so on.

This file is used to set many configuration parameters of the system, overclocking being one of them. The 'config.txt' is read by the GPU before the ARM core is initialised. Open 'config.txt' file using the command:

```
$ sudo nano /boot/
config.txt
```

And add the following lines (refer Fig. 3):

```
arm_freq=800
sdram_freq=500
```

Press 'Ctrl+o' to save and 'Ctrl+x'